Jacobian Descent For Multi-Objective Optimization

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Joint work with Pierre Quinton

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GitHub
github.com/TorchJD/torchjd

TorchJD
torchjd.org

arXiv
arxiv.org/pdf/2406.16232

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Theory

Partial order

Define the partial order $(\mathbb{R}^m, \preceq)$, for $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^m$, as

$$\begin{aligned} & \boldsymbol{u} \preceq \boldsymbol{v} & \\ \Leftrightarrow \quad & u_i \leq v_i & \forall i \in [m] \end{aligned}$$

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Examples:

- $\begin{bmatrix} 2 \\ 1 \end{bmatrix} \preceq \begin{bmatrix} 2 \\ 2 \end{bmatrix}$

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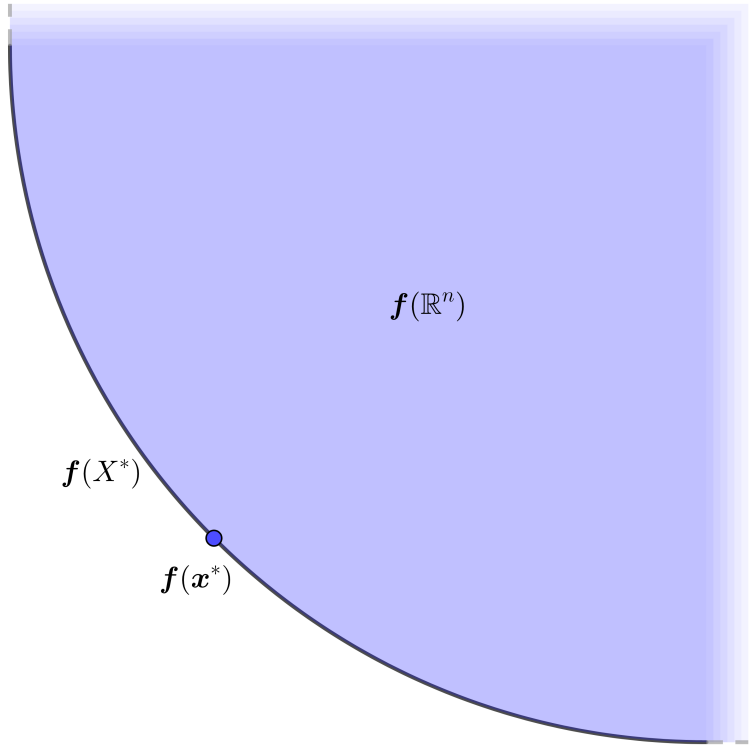
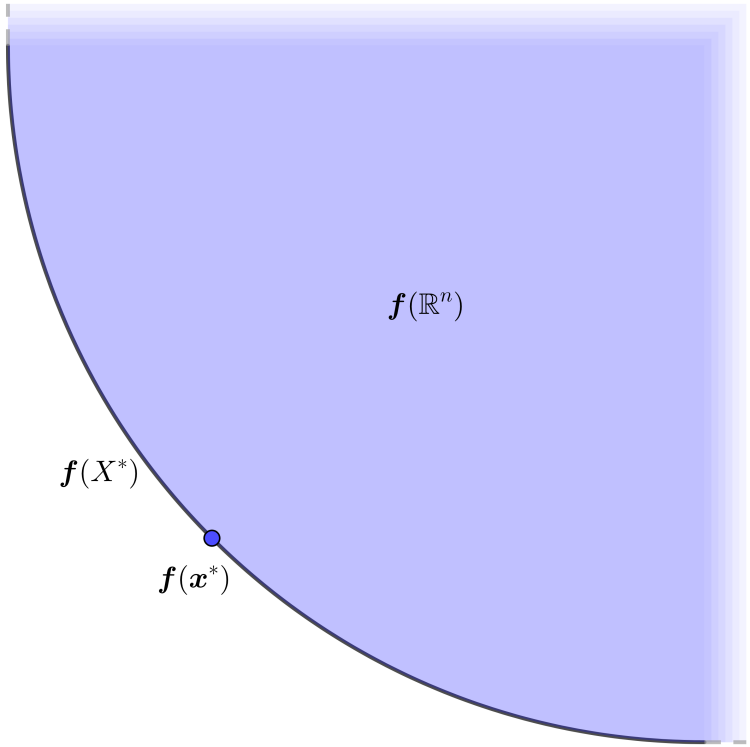
- $\begin{bmatrix} 2 \\ 1 \end{bmatrix} \preceq \begin{bmatrix} 2 \\ 2 \end{bmatrix}$
- $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ are not comparable with this partial order.

Pareto optimality



Pareto optimality

Pareto set: X^*



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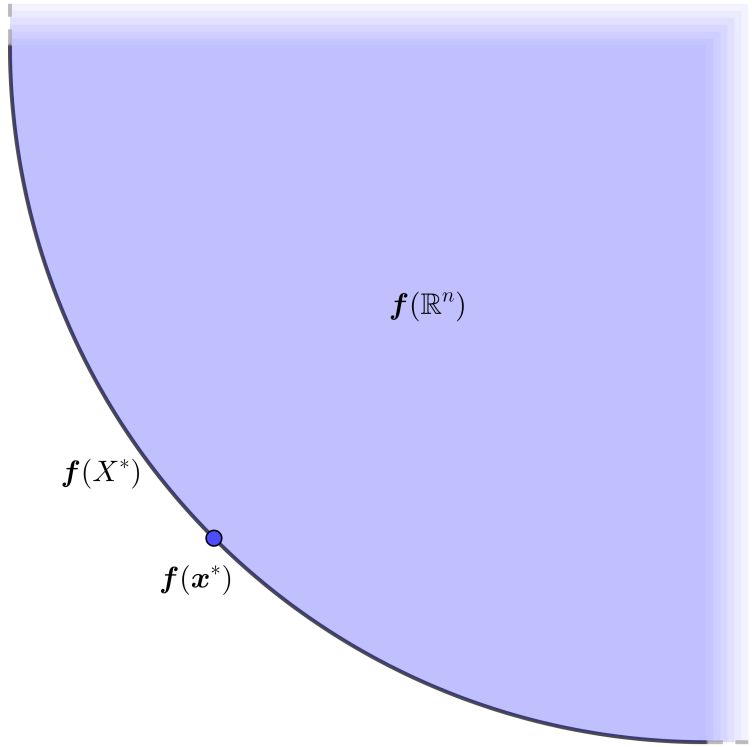

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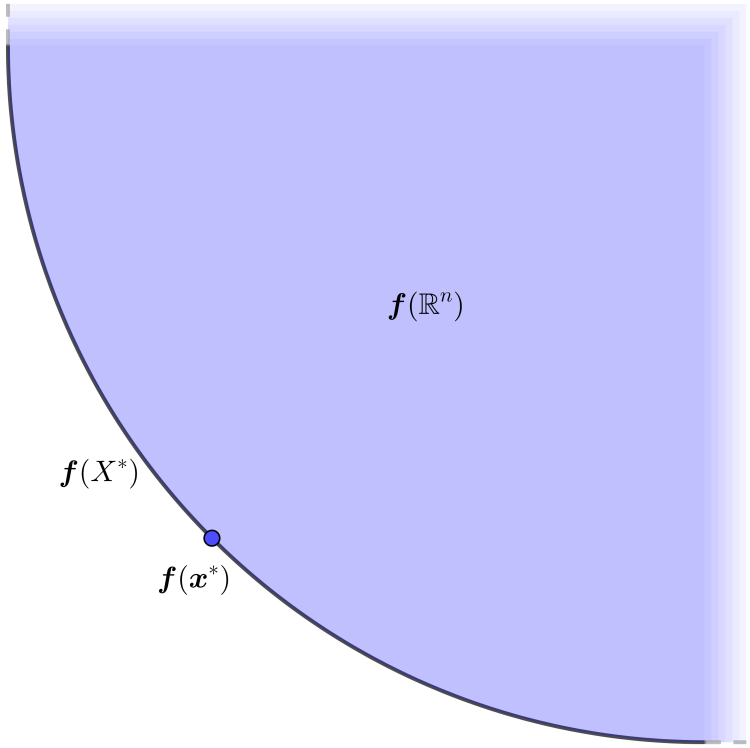

$$\boldsymbol{x}^* \in X^* \quad \Leftrightarrow \quad \forall \boldsymbol{y} \in \mathbb{R}^n,\ \boldsymbol{f}(\boldsymbol{y}) \preceq \boldsymbol{f}(\boldsymbol{x}^*) \rightarrow \boldsymbol{f}(\boldsymbol{y}) = \boldsymbol{f}(\boldsymbol{x}^*)$$

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$$\boldsymbol{f}(\boldsymbol{x} + \boldsymbol{y}) \approx \boldsymbol{f}(\boldsymbol{x}) + \mathcal{J}\boldsymbol{f}(\boldsymbol{x}) \cdot \boldsymbol{y}$$

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We want $\boldsymbol{f}(\boldsymbol{x} + \boldsymbol{y}) \preceq \boldsymbol{f}(\boldsymbol{x})$. We select $\boldsymbol{y} = -\eta \mathcal{A}(\mathcal{J}\boldsymbol{f}(\boldsymbol{x})) \in \mathbb{R}^n$.

Jacobian descent

The mapping $\mathcal{A} : \mathbb{R}^{m \times n} \to \mathbb{R}^n$ is called an **aggregator**, it parameterizes **Jacobian descent**:

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Algorithm 1: Jacobian descent with aggregator $\mathcal{A}$

Input: $\boldsymbol{x} \in \mathbb{R}^n$, $0 < \eta$, $T \in \mathbb{N}$, $\mathcal{A} : \mathbb{R}^{m \times n} \to \mathbb{R}^n$

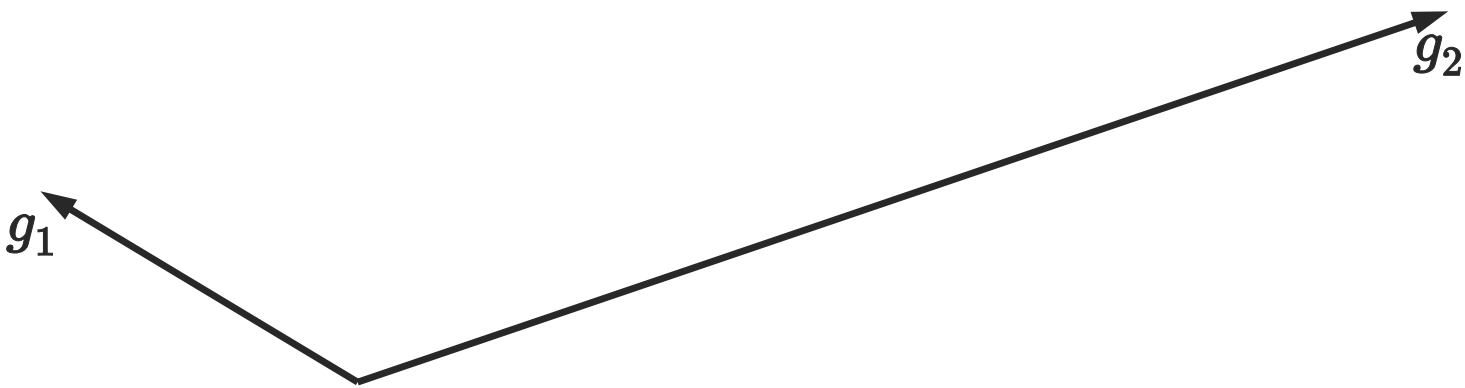
for $t \leftarrow 1$ **to** T **do**

$\quad \boldsymbol{x} \leftarrow \boldsymbol{x} - \eta \mathcal{A}(\mathcal{J}\boldsymbol{f}(\boldsymbol{x}))$

Output: $\boldsymbol{x}$

Unconflicting projection of gradients (UPGrad)

$$J = \begin{bmatrix} g_1^\top \\ g_2^\top \end{bmatrix}$$



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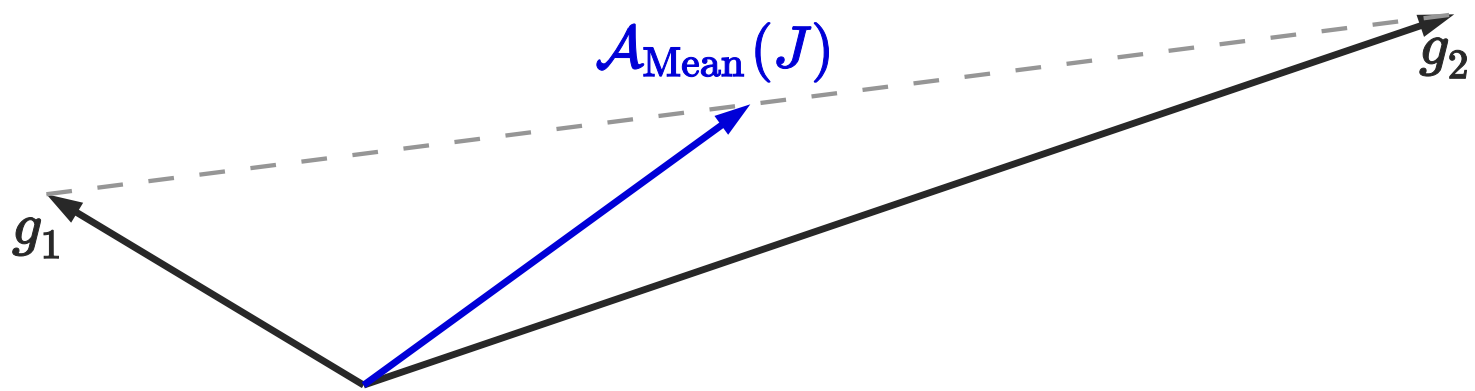

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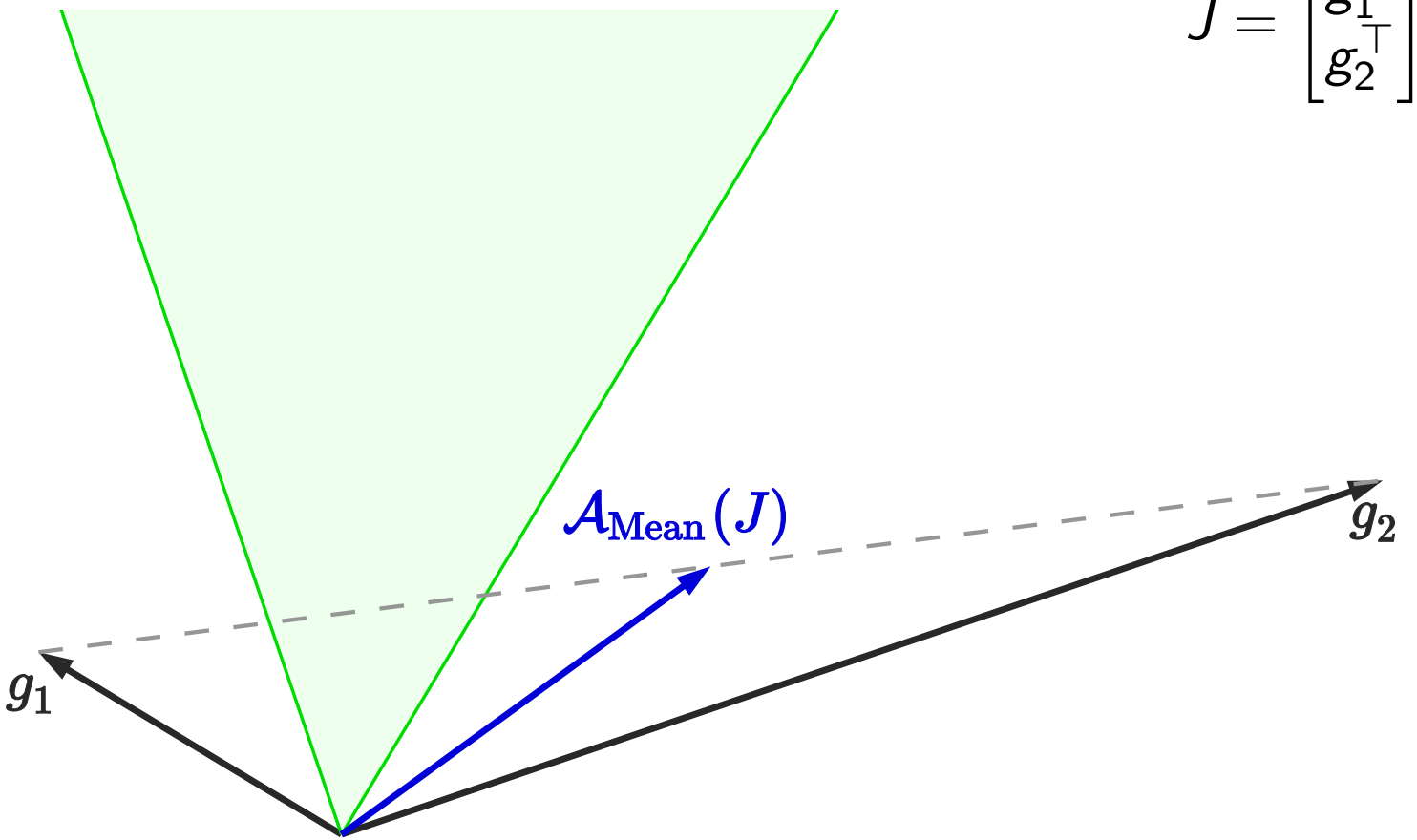

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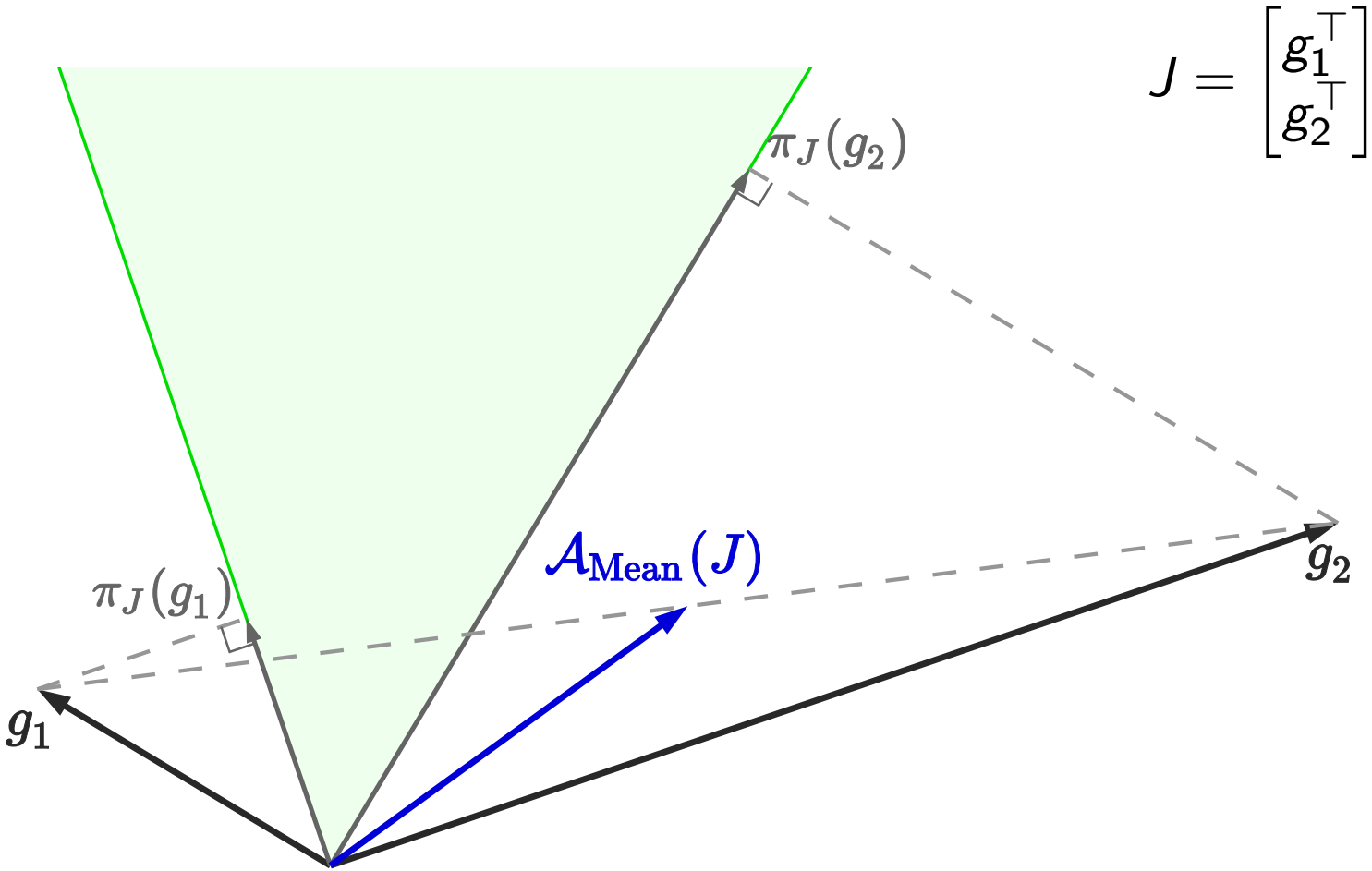

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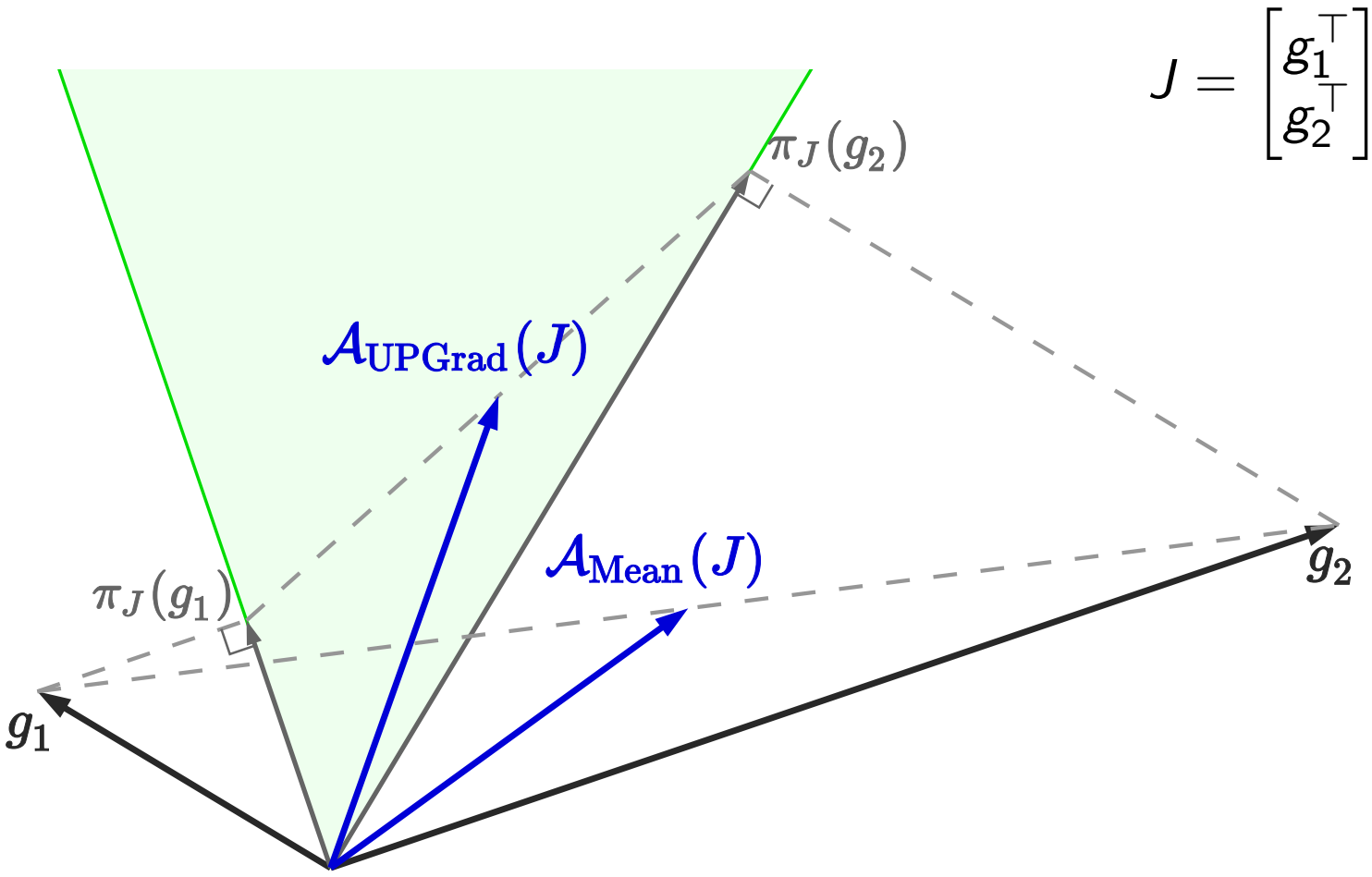


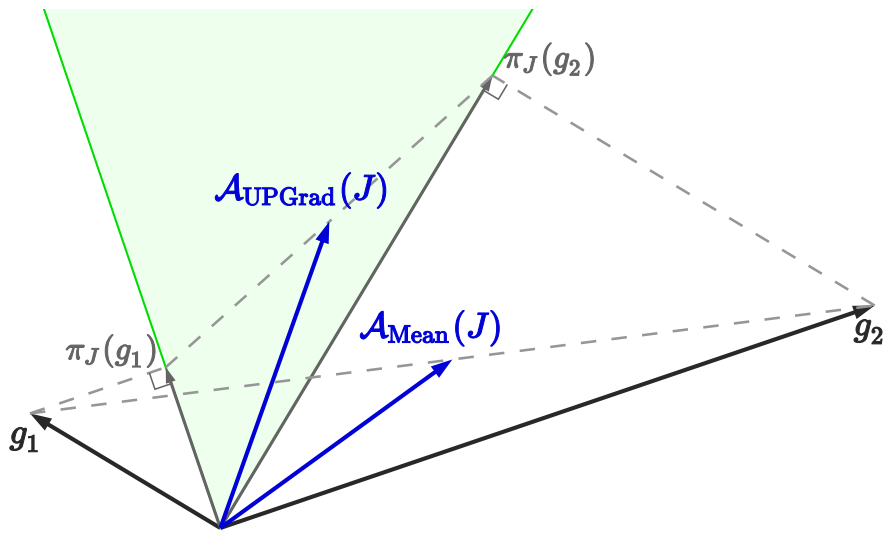
Formal definition of UPGrad

Given $J \in \mathbb{R}^{m \times n}$, let C_J be the cone $\{\boldsymbol{y} \in \mathbb{R}^n : \boldsymbol{0} \preceq J\boldsymbol{y}\}$. $\mathcal{A}$ is non-conflicting if for all $J \in \mathbb{R}^{m \times n}$, $\mathcal{A}(J) \in C_J$.

For any $\boldsymbol{x} \in \mathbb{R}^n$, the projection of $\boldsymbol{x}$ onto C_J is

$$\pi_J(\boldsymbol{x}) = \underset{\boldsymbol{y} \in C_J}{\arg\min} \|\boldsymbol{x} - \boldsymbol{y}\|^2$$

$$\mathcal{A}_{\text{UPGrad}}(J) = \frac{1}{m} \sum_{i \in [m]} \pi_J(g_1)$$



Applications and experimentation

Optimization trajectories

Goal: Minimize some convex quadratic function $\boldsymbol{f} : \mathbb{R}^2 \to \mathbb{R}^2$, with conflicting gradients.

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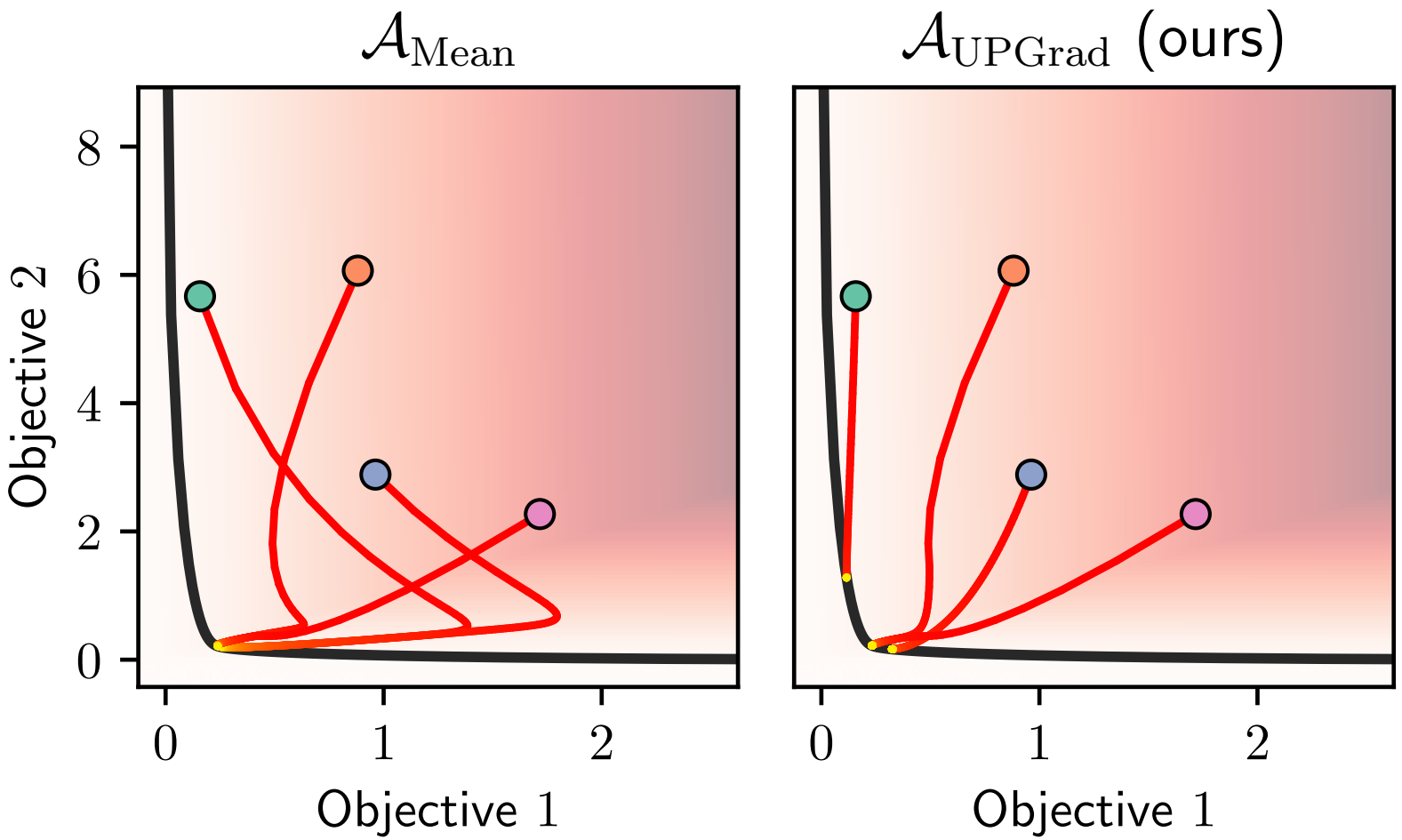

Optimization trajectories: code

```
from torchjd import autojac
from torchjd.aggregation import UPGrad

x = tensor([0., 0.])  # initial value of x

for i in range(n_iters):
    y = f(x)  # shape: [2]
    autojac.backward(y)
    # x now has a .jac field
    autojac.jac_to_grad([x], UPGrad())
    # x now has a .grad field
    optimizer.step()
    optimizer.zero_grad()
```

Full project available at github.com/TorchJD/trajectories

Per-sample JD

Each element of the batch has its own loss.

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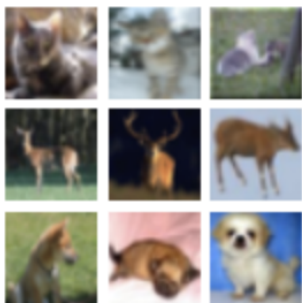
```
from torchjd import autojac
from torchjd.aggregation import UPGrad

batch_size = 16
model = Linear(10, 1)

for x, y in dataloader:
  z = model(x).squeeze(dim=1)  # shape: [16]
  losses = loss_fn(z, y)  # shape: [16]
  autojac.backward(losses)
  # parameters now have a .jac field
  autojac.jac_to_grad(model.parameters(), UPGrad())
  # parameters now have a .grad field
  optimizer.step()
  optimizer.zero_grad()
```

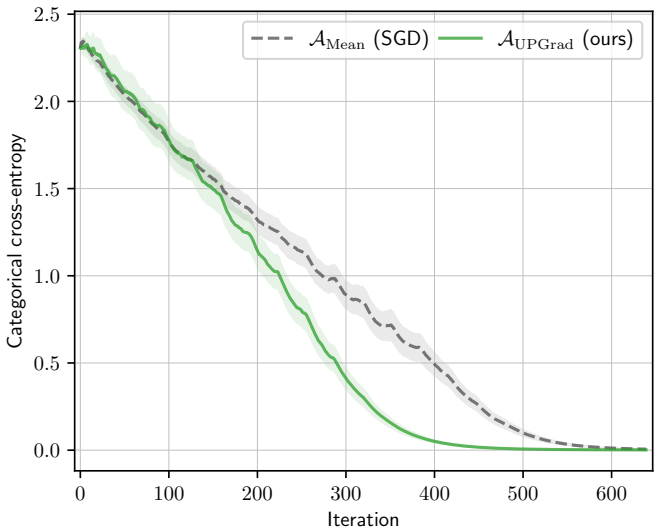
Full code example available at torchjd.org/latest/examples/iwrm.

Per-sample JD: results

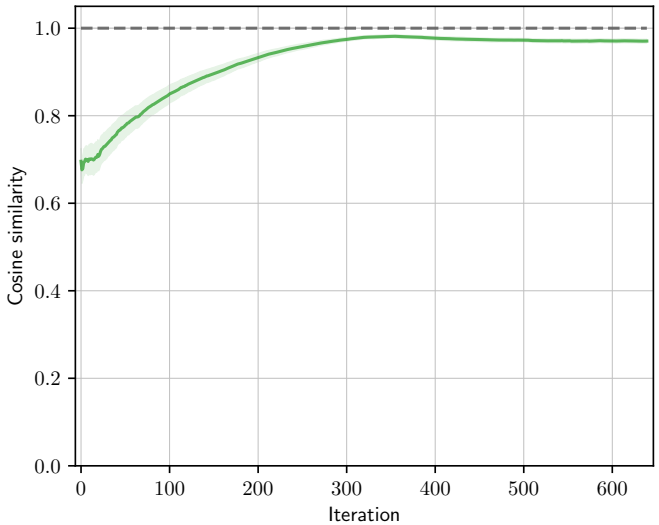
CIFAR-10



Training Loss



Update Similarity



Per-sample JD: results

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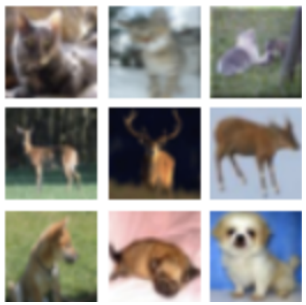
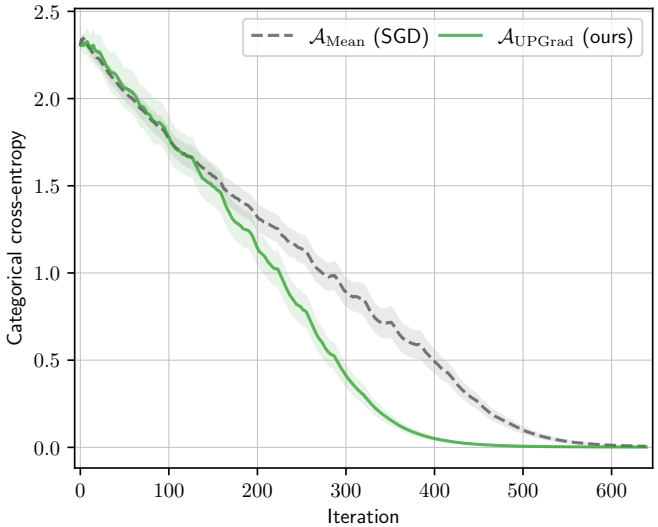
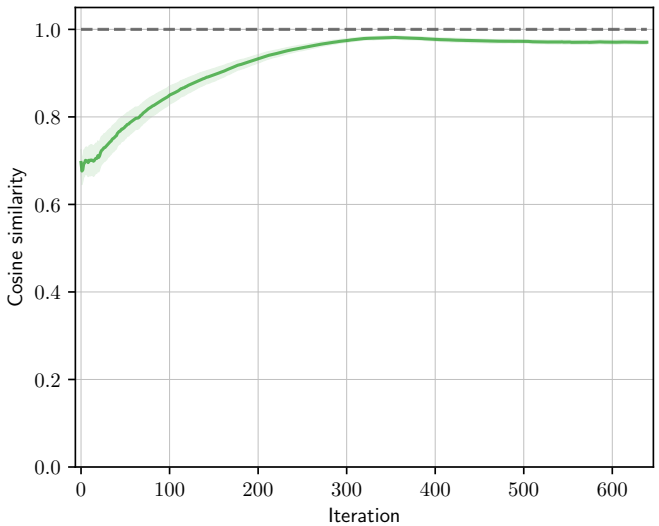
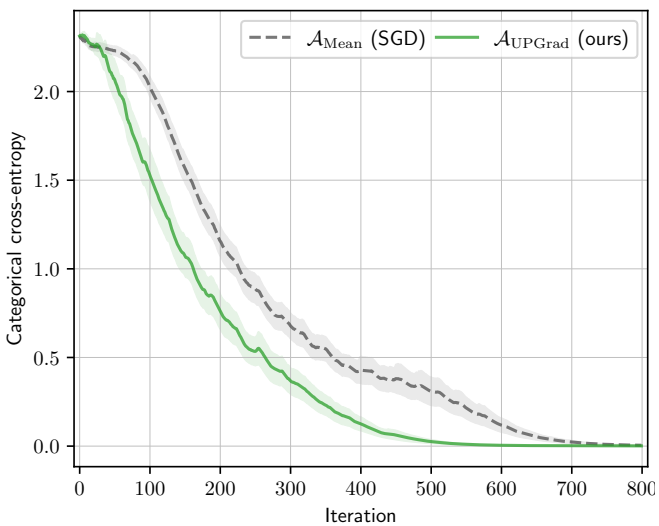
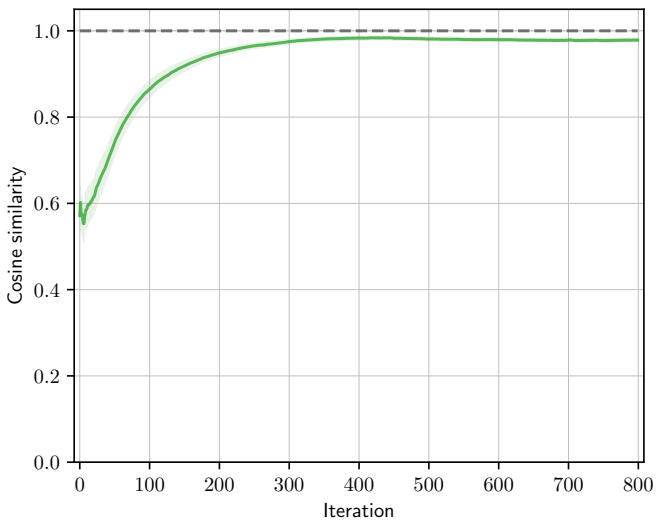






SVHN





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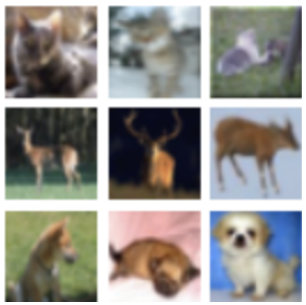

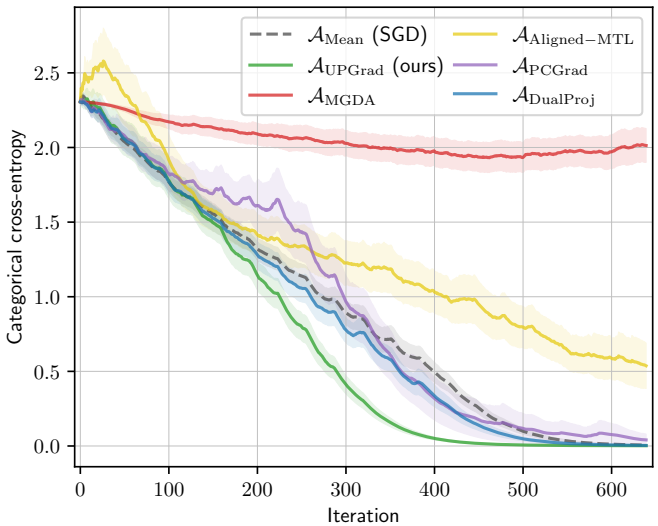

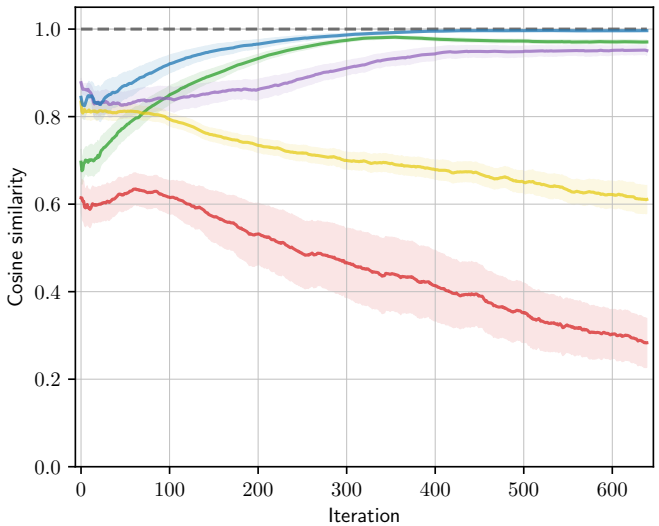

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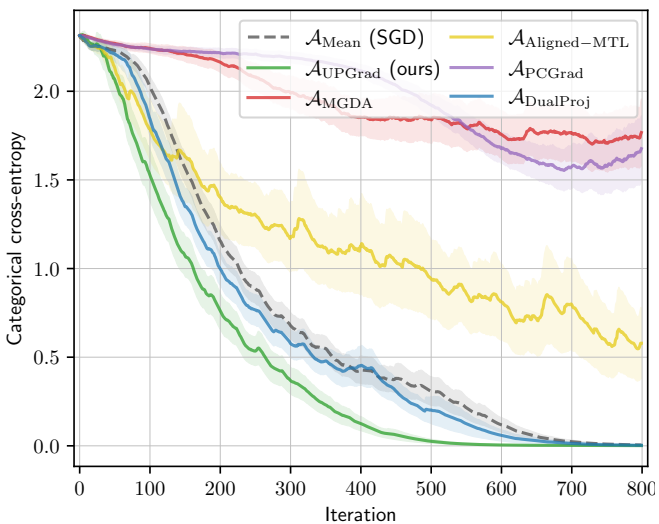

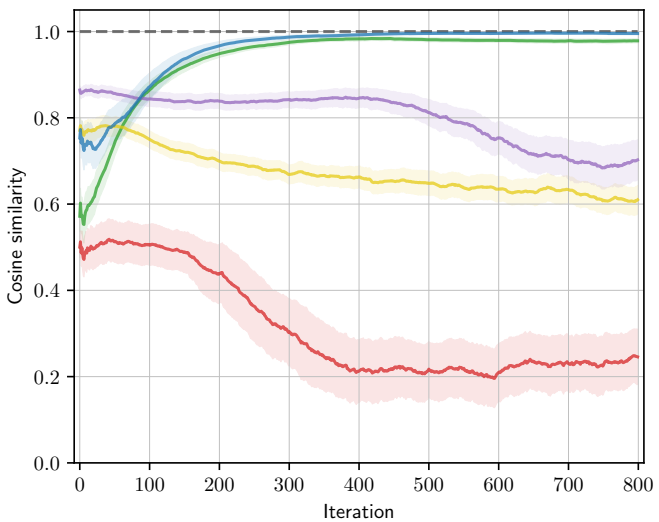

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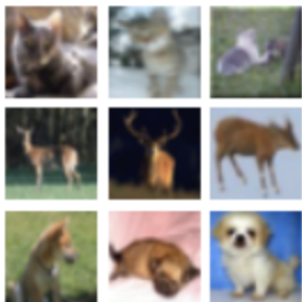

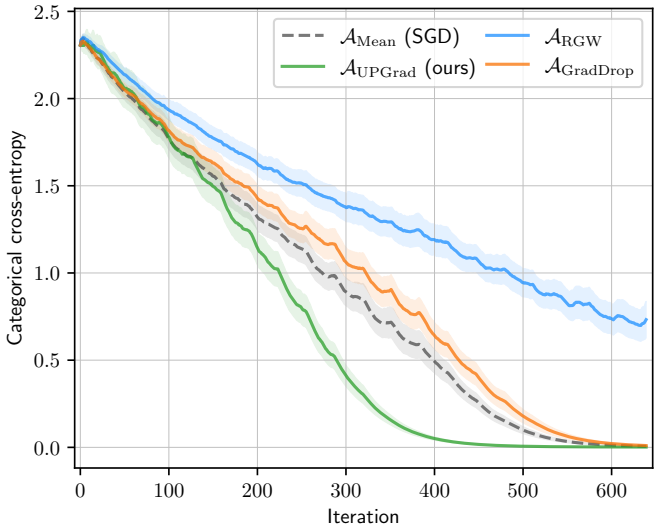

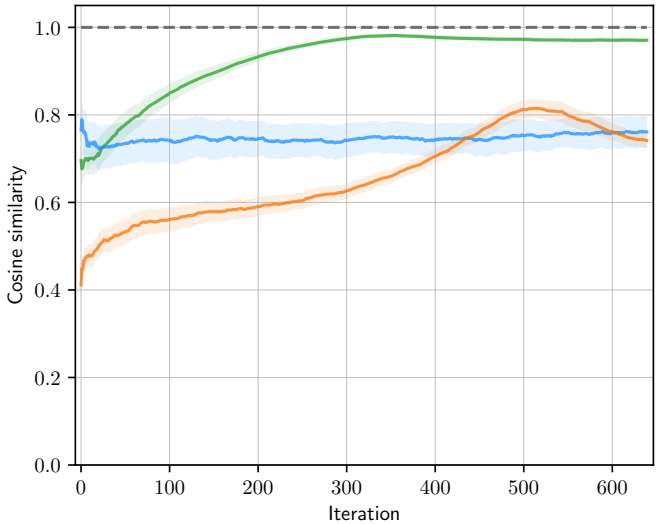

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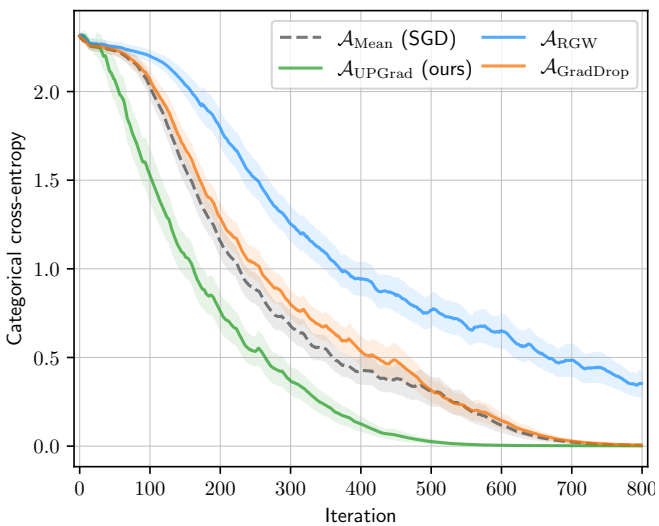

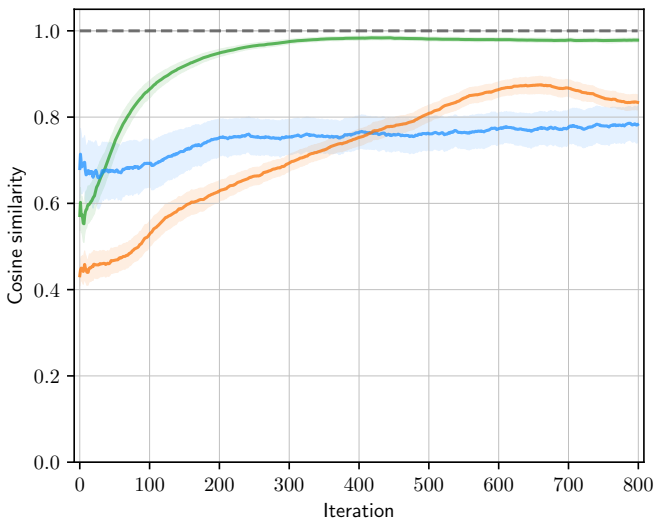

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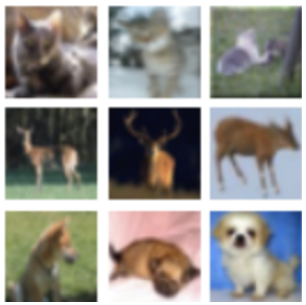
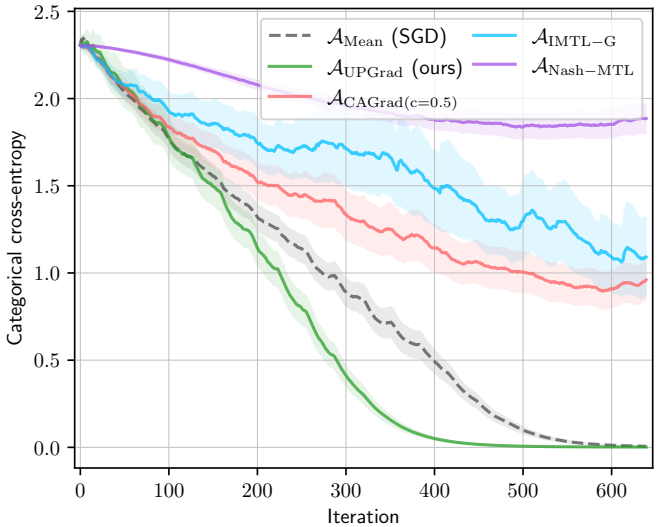
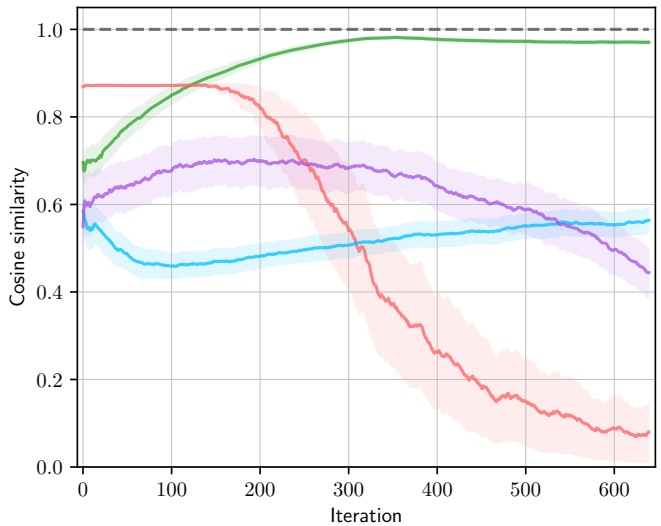
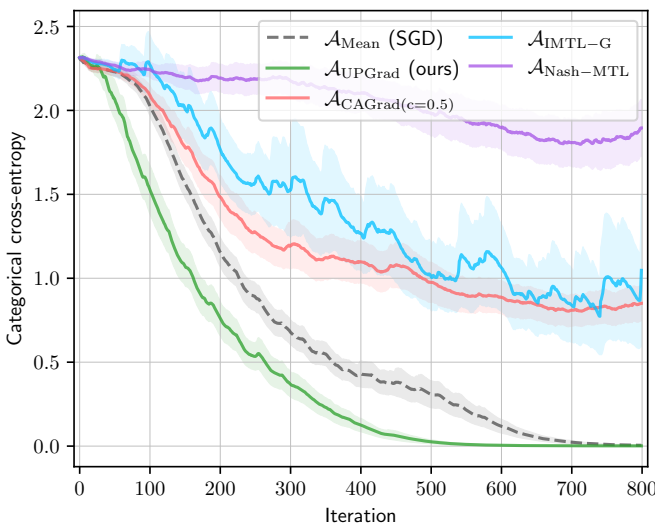
CIFAR-10

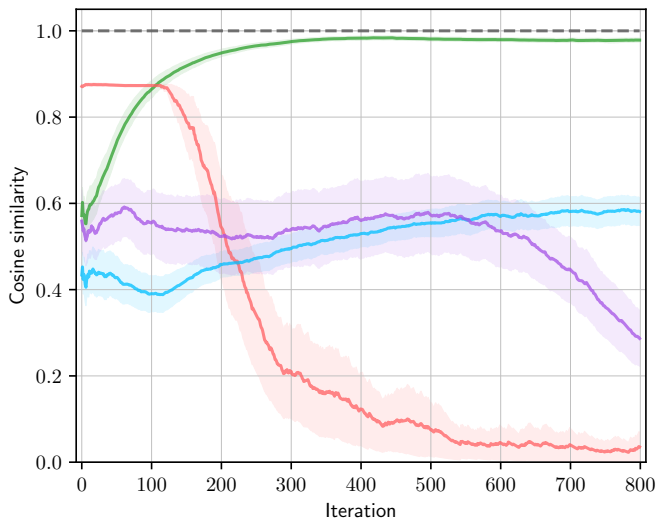




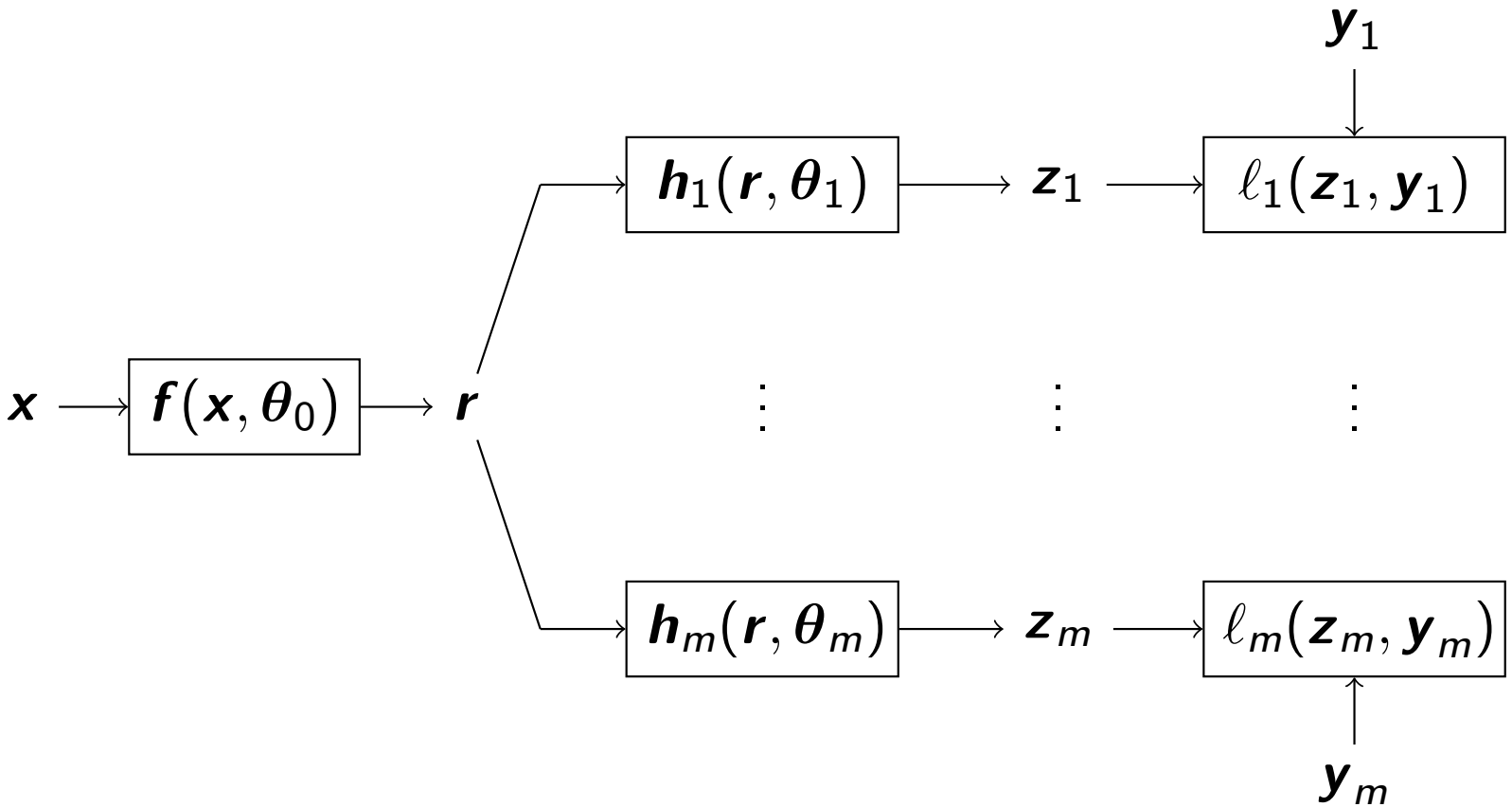


SVHN





Multi-task learning



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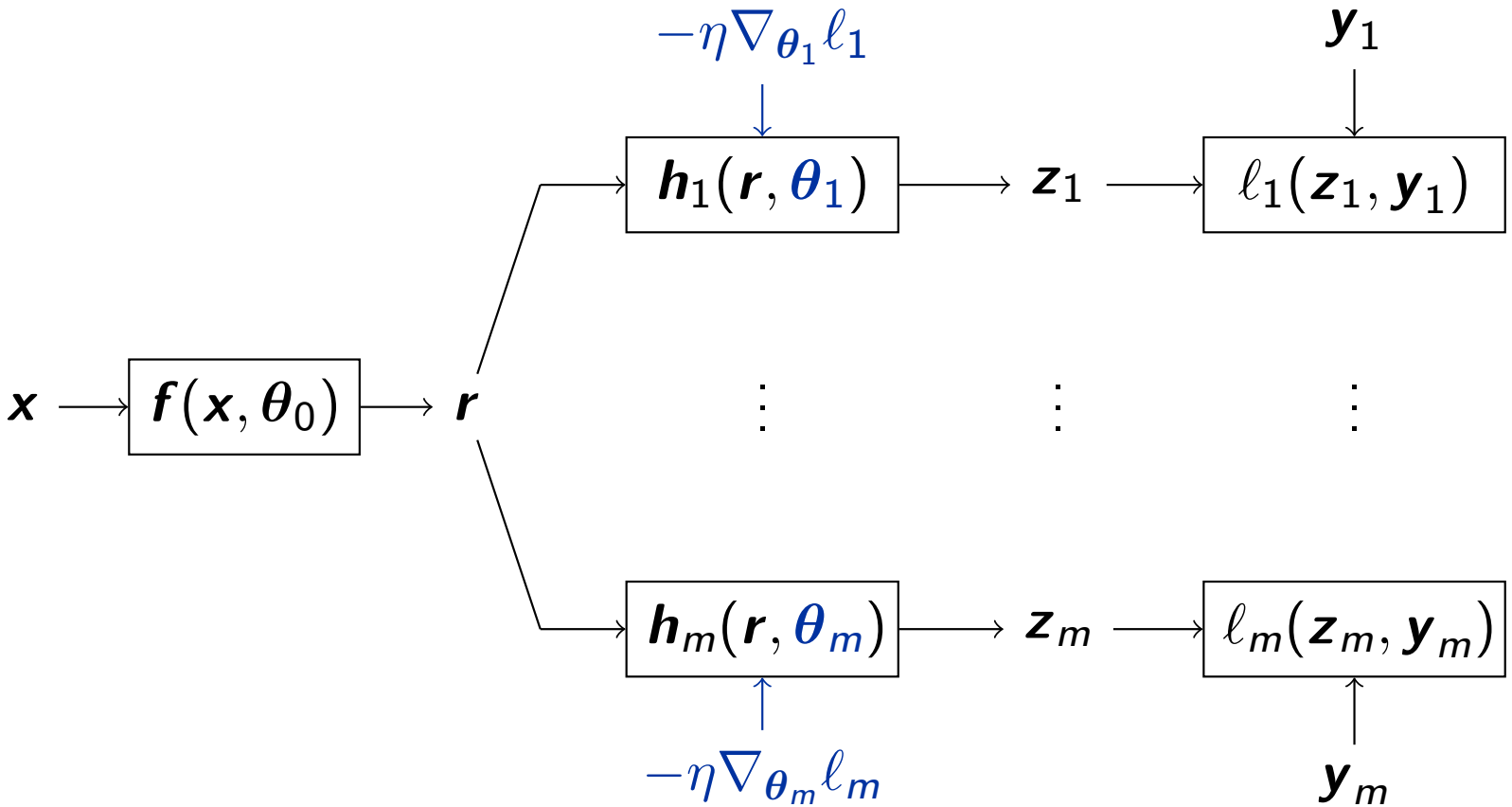

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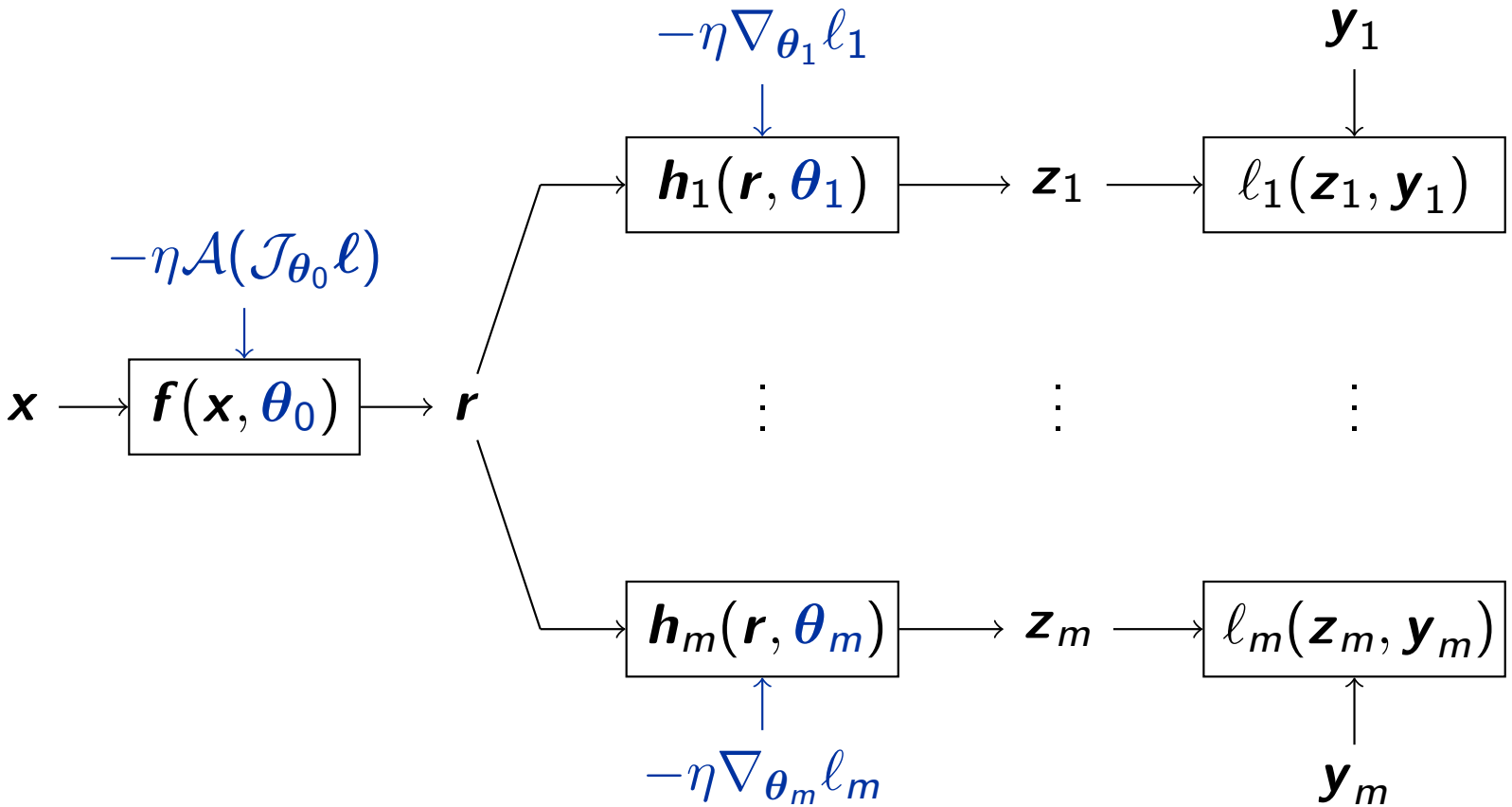

Multi-task learning: code

```
from torchjd import autojac
from torchjd.aggregation import UPGrad

f = Linear(10, 5)
h1 = Linear(5, 1)
h2 = Linear(5, 1)
for x, y1, y2 in dataloader:
  r = f(x)
  z1, z2 = h1(r), h2(r)
  l1, l2 = loss_fn(z1, y1), loss_fn(z2, y2)
  autojac.mtl_backward([l1, l2], features=r)
  # head parameters now have a .grad field
  # backbone parameters now have a .jac field
  autojac.jac_to_grad(f.parameters(), UPGrad())
  # All parameters now have a .grad field
  optimizer.step()
  optimizer.zero_grad()
```

Full code example available at torchjd.org/latest/examples/mtl.

Conclusion

Many problems are multi-objective.

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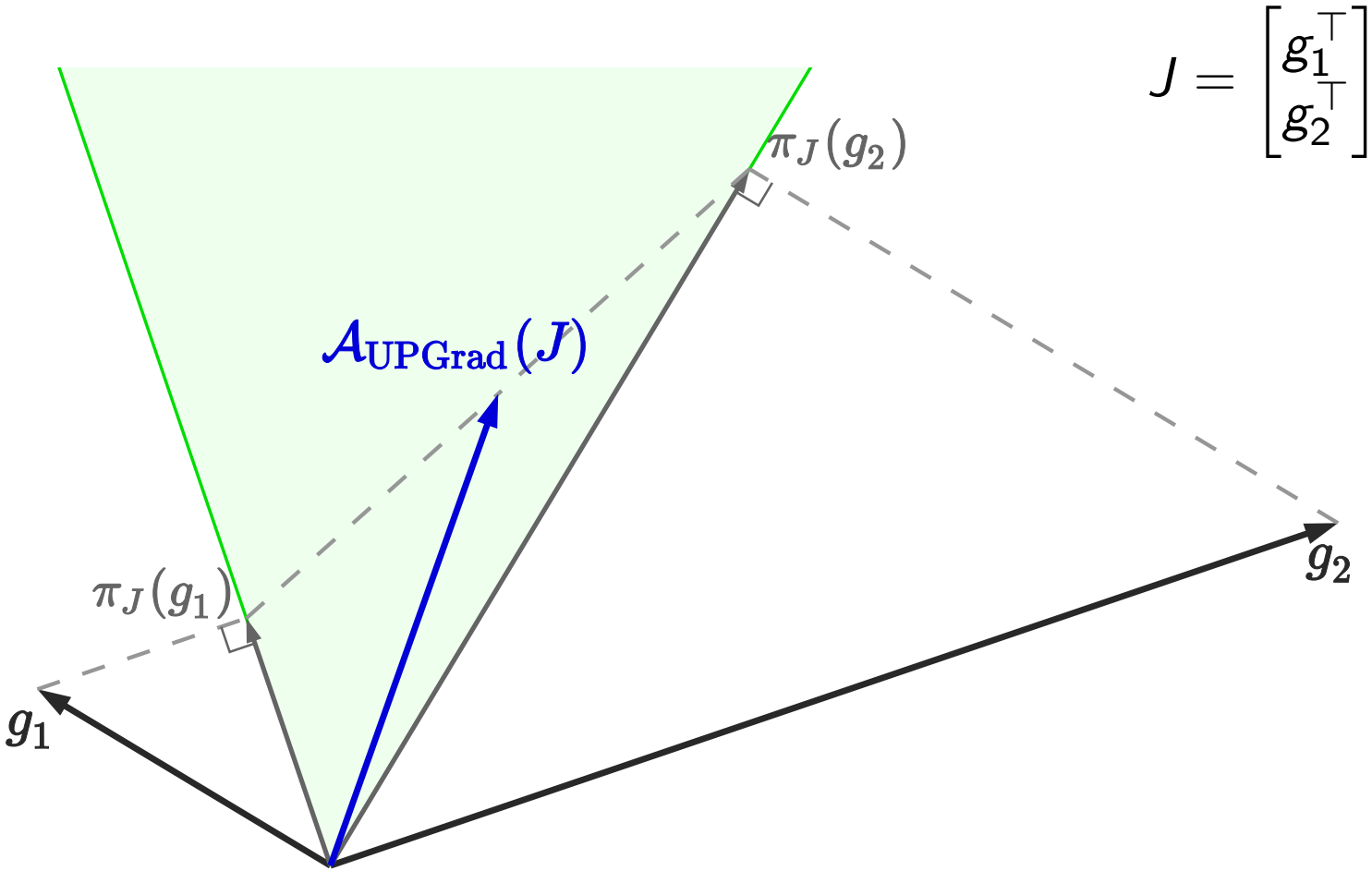
Try to compute the inner products between your gradients.

Appendix

Other aggregators



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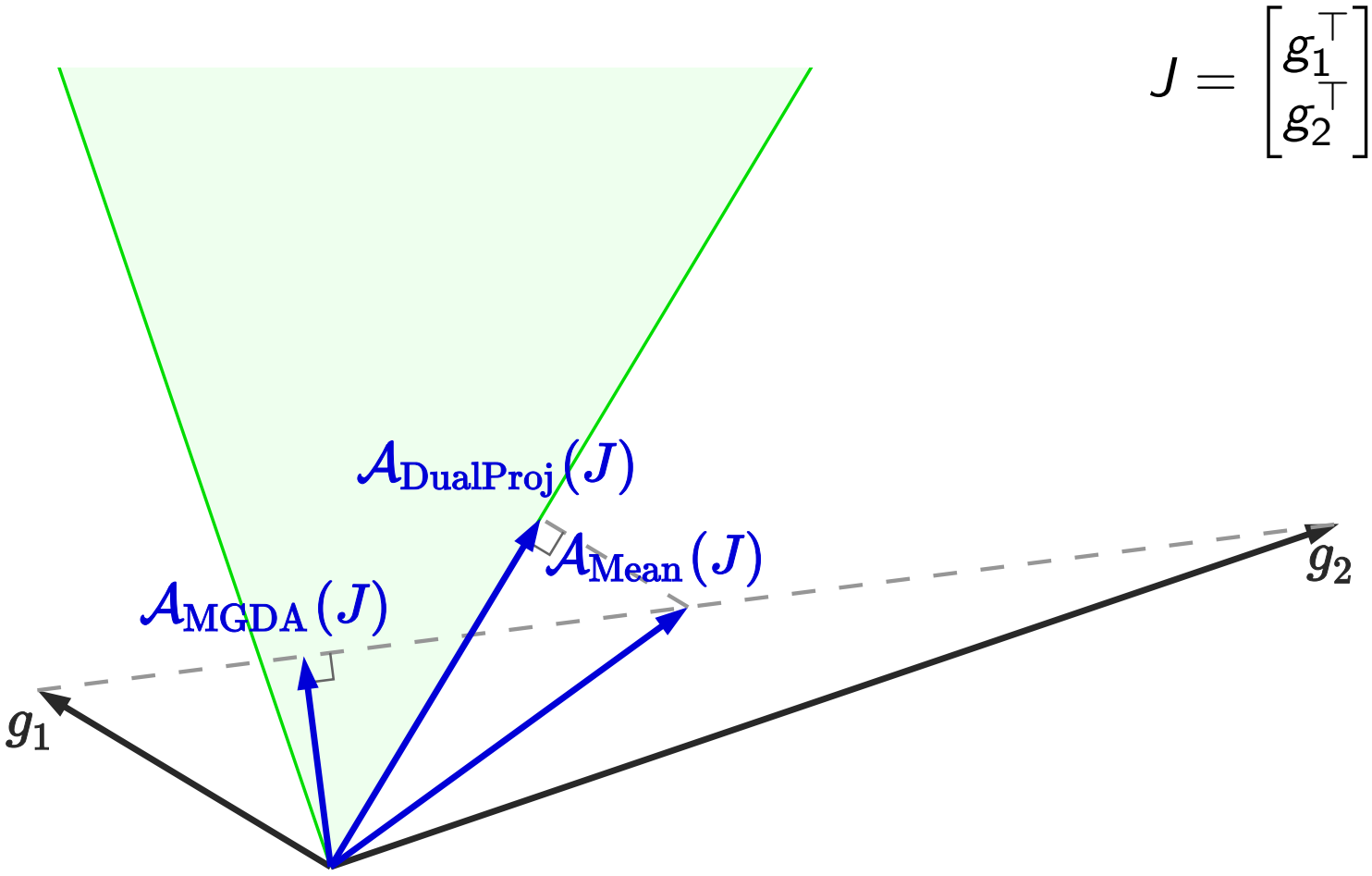
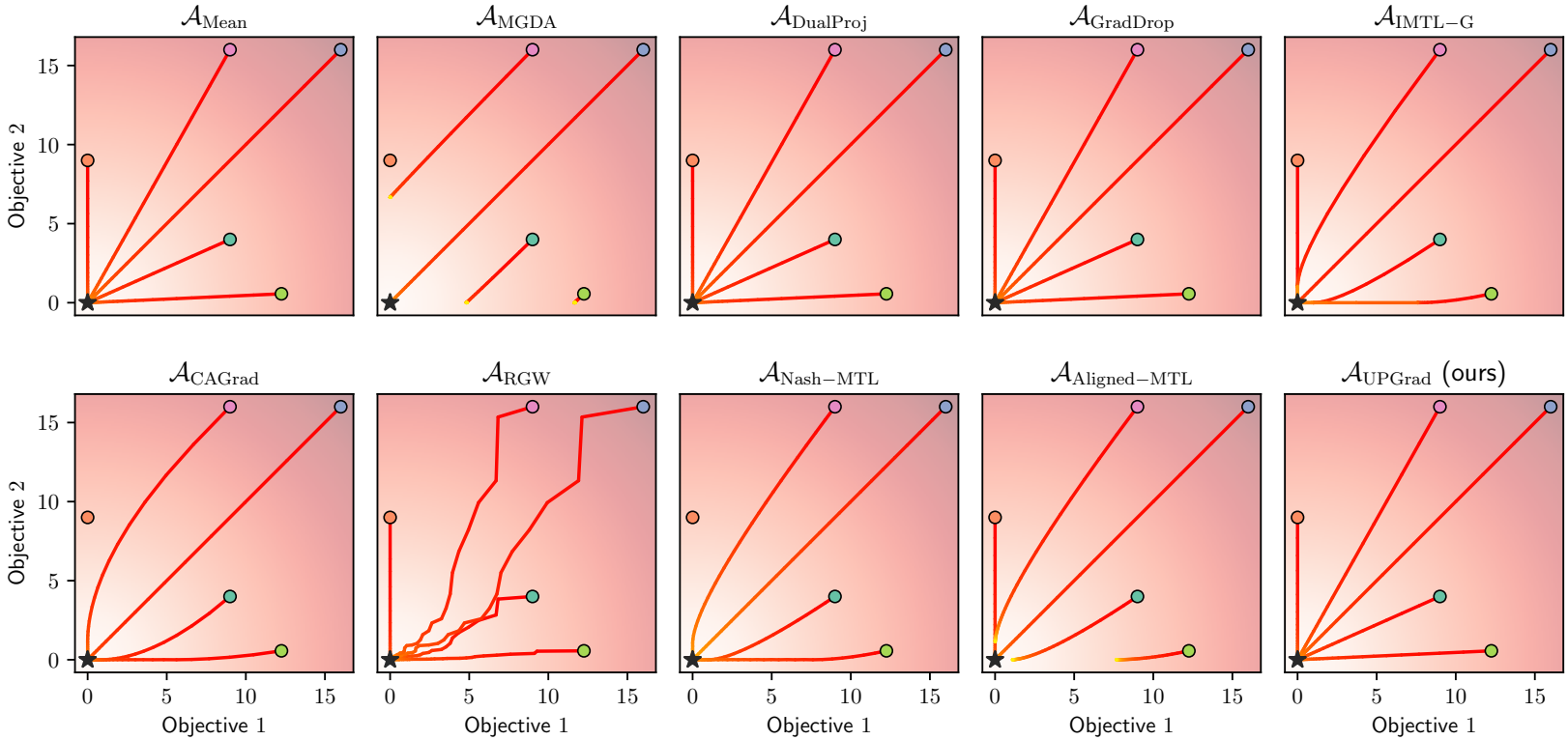


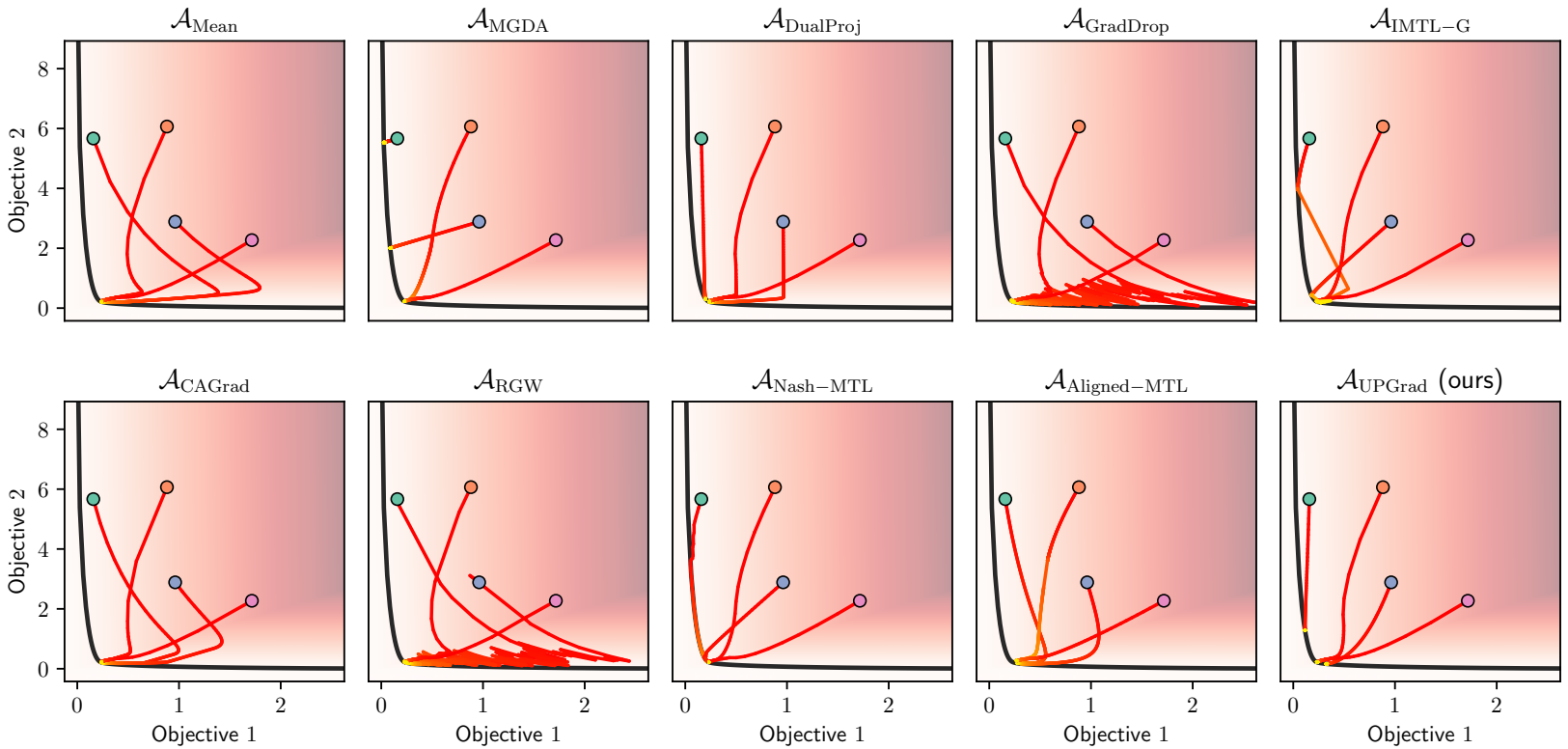
Element-wise quadratic function trajectories

Minimize $\boldsymbol{f}$: $\begin{bmatrix} x \\ y \end{bmatrix} \mapsto \begin{bmatrix} x^2 \\ y^2 \end{bmatrix}$. It's Jacobian at $\begin{bmatrix} x \\ y \end{bmatrix}$ is $\begin{bmatrix} 2x & 0 \\ 0 & 2y \end{bmatrix}$.



Convex quadratic form trajectories

Some convex quadratic function $\boldsymbol{f} : \mathbb{R}^2 \to \mathbb{R}^2$, with conflicting gradients.



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Applicable to most other aggregators from the literature.